Data Flow Diagram and Data Movement

Trip Planner — Supporting Project Document

Context-Level View

External actor: Traveller. Main system: Trip Planner application. External services: Groq AI service for generation and Supabase PostgreSQL for persistence. The traveller supplies preferences and receives generated content or saved-trip results.

Level 1 — Generation Flow

• 1. Traveller enters destination and preferences in the frontend.

• 2. Frontend sends a request to the FastAPI backend.

• 3. Backend validates the request and builds a prompt.

• 4. Backend sends the prompt to Groq.

• 5. Groq returns generated planning text.

• 6. Backend sends the result to the frontend.

• 7. Frontend displays itinerary, budget and packing content.

Level 1 — Save and Retrieve Flow

• 1. Traveller chooses Save Complete Trip.

• 2. Frontend sends trip fields and generated outputs to the backend.

• 3. Backend writes the record to Supabase PostgreSQL.

• 4. Backend returns success or a safe error message.

• 5. My Trips requests saved records.

• 6. Backend reads records and returns them for display.

Data Entities

A saved record includes an identifier, destination, duration, traveller count, budget, travel style, interests, itinerary, budget breakdown, packing list and creation timestamp.

Security Notes

Provider credentials remain on the server. Public multi-user deployment requires authentication, authorization, user-specific access policies and rate limiting.